Section 01

Industrial Security (Bachelor)

Section 01 – Introduction to Industrial Security

IS Security

Introduction to Industrial Security

Lecture Notes

Department of Computer Science

Outline – Section 01

1 Cyber-Physical Foundations

2 Context and History

3 Threat Setting and Misconceptions

Learning Objectives

By the end of this section you will be able to

- Define Operational Technology (OT) and contrast it with Information Technology (IT).
- Identify the core components of an Industrial Control System (ICS).
- Explain why the CIA triad is re-ordered for industrial environments.
- Place industrial security in its historical and economic context.
- Recognise common misconceptions about ICS security.

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Cyber-Physical Foundations

Section 01 – Introduction to Industrial Security

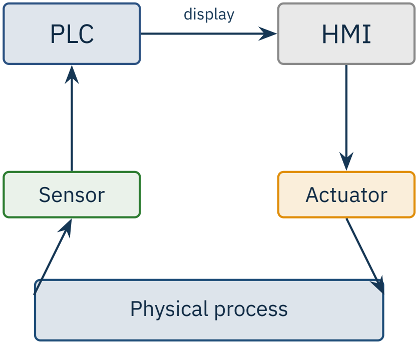
What is Industrial Security?

Definition

Industrial security is the protection of cyber-physical systems — computers that monitor and control machines in the physical world.

Sectors: power, water, oil & gas, manufacturing, transport, building automation.

A breach has consequences in *atoms*, not just in *bits*.



IT vs. OT — Two Different Worlds

Information Technology

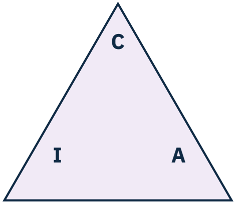
Conf. $\succ$ Integ. $\succ$ Avail.
Lifecycle: 3–5 years
Frequent patches
Soft real-time
Office environment
Failure cost: data loss

converge

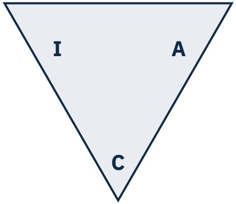
Operational Technology

Avail. $\succ$ Integ. $\succ$ Conf.
Lifecycle: 15–30 years
Patches require shutdown
Hard real-time, deterministic
Hostile environment
Failure cost: safety, lives

The CIA Triad — Inverted for OT



IT priorities



OT priorities

Remember

In OT, **availability** and **safety** dominate the CIA triad. A blocked alarm can blind an operator to a runaway reactor.

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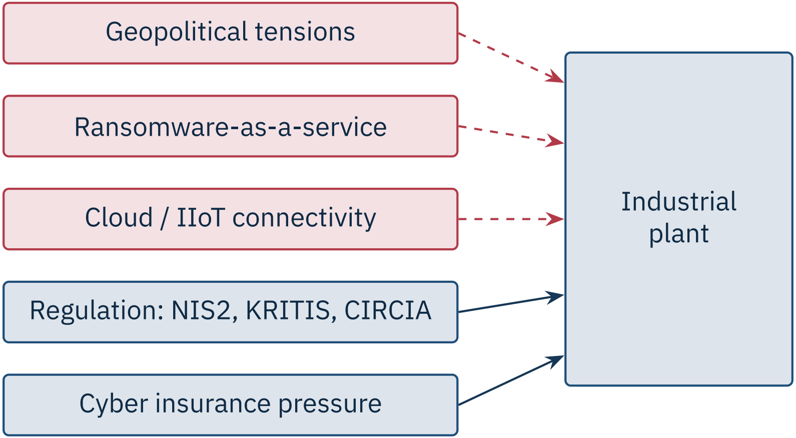
Context and History

A Brief History of Industrial Control

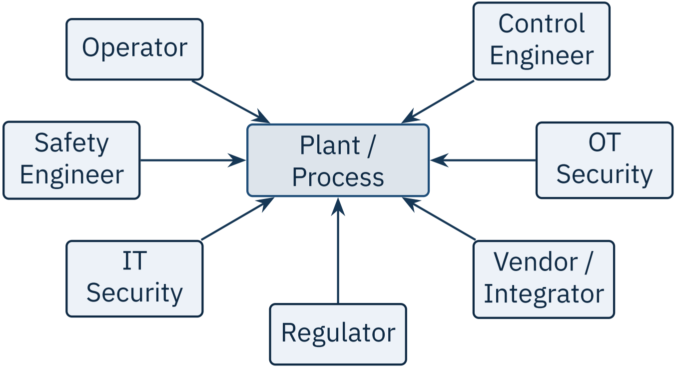
Each milestone made systems more capable – and more reachable.

Why Industrial Security Now?



Source: Dragos *Year in Review 2024*: 23 active threat groups tracked, ICS ransomware up 87% YoY | Munich Re *Cyber Insurance Risk Snapshot*, 2024.

Stakeholders



Industrial security is a team sport across roles that rarely sat together historically.

3

Threat Setting and Misconceptions

Common Misconceptions

“We’re air-gapped, so we’re safe.”

“Our protocols are obscure; nobody understands them.”

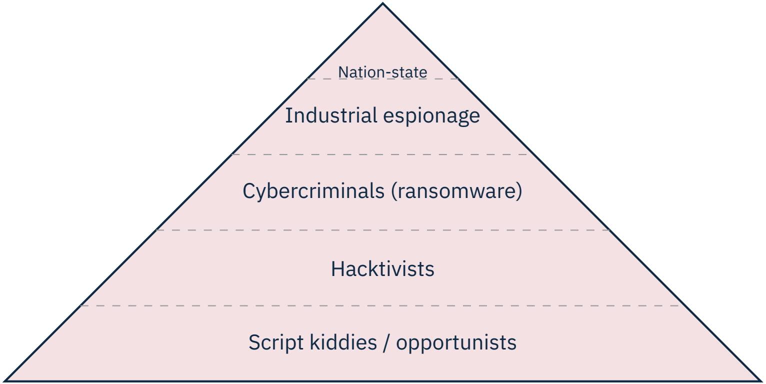
“We’ve never been attacked, so we’re fine.”

“IT tools will solve OT problems.”

“Patching solves everything.”

“Compliance equals security.”

Threat Pyramid – Who Targets ICS?



Capability rises as we move up; volume of actors falls.

The Cost of ICS Incidents

Colonial Pipeline	Norsk Hydro	Maersk (NotPetya)	Stuxnet
\$4.4M ransom 5-day outage	\$71M loss 6-month recovery	\$300M loss 10 days down	~1000 centrifuges multi-year program

Direct + indirect

Recovery costs typically exceed ransom by 5–10×. The lasting damage is reputation and regulatory.

Source: Colonial: FBI 2021; Norsk Hydro Q1 2019 report; Maersk Q2 2017 interim report; Stuxnet centrifuge count: D. Albright et al., ISIS, 2010.

Industry 4.0 – Convergence Pressure

PLC / DCS → Edge gateway → Cloud analytics → ML / digital twin

New attack surface

Every cloud connector is a new ingress – and a new authentication challenge for legacy field devices.

Career Paths in Industrial Security

- **OT Security Engineer** – design, deploy, monitor controls; bridges IT and OT
- **Control Engineer with security focus** – starts from process side
- **ICS Penetration Tester** – offensive; specialised employers
- **Compliance / Audit** – IEC 62443, NIS2 specialist
- **Incident Responder (OT)** – on-call, often consulting-side (Dragos, Mandiant)
- **Researcher** – vendor / academic; vulnerabilities, threat intel

Demand

Headhunter reports consistently rank OT security in the top three hardest-to-fill cyber roles globally.

Section Roadmap

- ICS & SCADA fundamentals
- Network architectures
- Industrial protocols
- Threat landscape
- Vulnerabilities & assessment
- Risk management
- Defense in depth
- Standards & governance
- Incident response

Summary – Section 01

Key Takeaway: Industrial security is different from IT security

- OT differs fundamentally from IT in goals, lifecycle, and tolerance for failure.
- Availability and safety dominate the CIA triad in industrial contexts.
- Adversaries today span from script kiddies to nation-state APTs.
- The next sections build the toolbox: components, networks, protocols, defenses.

References – Section 01

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End of Section 01

Introduction to Industrial Security

Questions and discussion

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